



SCHOOL OF COMPUTER TECHNOLOGY

AASD 4007
Design Thinking for A.I. Solutions
Session 3

Reza Moslemi, Ph.D., P.Eng.

Session 3



- Exercise: Interview practice and expert Interviews (In-class)
- A3: Personas (In-class)
- Design Thinking Exercises: (In-class)
 - Empathy map
 - As-Is scenario map
 - Needs statements
 - AI intents
 - AI big idea vignettes
 - Debriefing to the class
- A4: To-Be-Scenarios & Data Sources (Homework)

Exercise: Interview practice and expert Interviews



Purpose of the Interview?

Demonstrate the research interview process:

1. Research questions
2. Interview Questions
3. Long list of Interview Questions
4. Themes to discuss in interview
5. Fit script into interview constraints

Demonstrate Teamwork

Gain Expertise

Answer Project Specific Questions

Help you do your own Research Interviews better

Exercise: Interview practice and expert Interviews



In-class Exercise 1 = 5% of grade

Mode: in group

Submission: on Brightspace as a MURAL link

Time: 45 min

This exercise is based on Assignment 2

- Round 1: Combine all the interview questions and select 3 top questions
 - Enter questions in MURAL
- Round 2: Conduct interview with Expert in your groups
 - Practice Interview with the selected questions
(break into two sub-groups and take turns to practice the roles of interviewer and participant)
 - Enter insights from Expert interview in MURAL
- Demo your Research/Interview questions in class

Assignment 3: In-class



Personas

Design Thinking Exercises



In-class Exercise 2= 10% of grade

Mode: in group

Submission: on Brightspace as a MURAL link

Time: 75min

This exercise is based on Assignment 3

- Empathy map
- As-Is scenario map
- Needs statements
- AI intents
- AI big idea vignettes
- Debriefing to the class

Empathy Map



<https://www.ibm.com/design/thinking/page/toolkit/activity/empathy-map>

VIDEO [01:22]

1. Pick one PRIMARY PERSONA from among your groups' A3 Personas
2. Fill Empathy Map through Primary Persona's perspective – their Actions (Do, Say, See, Hear), Thoughts(Think, Feel), Pains, & Gains

GROUP ACTIVITY (15 min)

Empathy Map

Sub-space: Transit

EMPATHY MAP



As-Is-Scenario Map



<https://www.ibm.com/design/thinking/page/toolkit/activity/as-is-scenario-map>

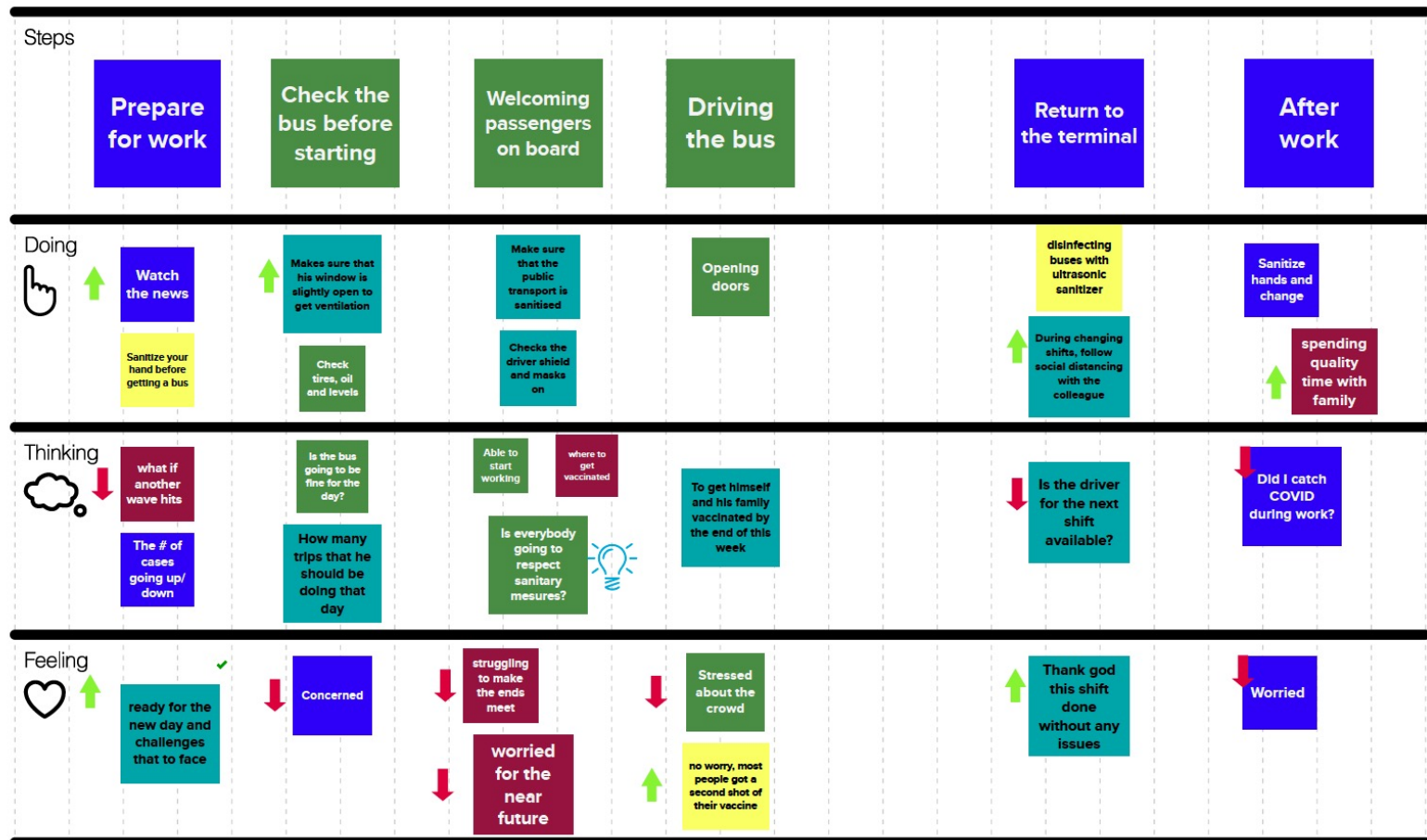
VIDEO [01:15]

1. Brainstorm individually your Persona's CURRENT EXPERIENCE. What are they Doing, Thinking, Feeling
2. Use timeline to order columns by experience phases. Name the columns /phases steps.
3. Identify highs & lows, positive & negative experiences for your user. Look for opportunities to help them.

GROUP ACTIVITY (15 min)

As-Is-Scenario Map

SCENARIO MAP



Needs Statements



<https://www.ibm.com/design/thinking/page/toolkit/activity/needs-statements>

VIDEO [01:41]

1. Individually brainstorm user NEEDS and BENEFITS. Avoid technical details of your solution at this point.
2. With your group, cluster similar ideas & discuss.
3. Write a few needs statements in the format: [Our users] need a way to [address this need] so they can [benefit in this way.]

GROUP ACTIVITY (15 min)

Needs Statements

NEEDS STATEMENT [Our users] need a way to [address this need] so they can [benefit in this way]

Bob the bus driver needs a way to make sure the bus is sanitized frequently so that they can ensure everyone feel safe to take the bus.

Bob needs to make sure that the bus is getting disinfected regularly

Bob needs to make sure the bus that he is driving is clean enough.

Need to use a bus which is sanitized

Need regular and frequent sanitization of the bus

Bob needs the public transport to be sanitised frequently

Bob the bus driver needs a way to get himself enthusiastic so he can continue helping people during the pandemic

Bob has to attend camps and sessions to keep himself updated for latest protocols and updates

Bob needs to be emotionally ready to drive.

Bob has to make sure that the next shift crew is ready

Bob needs more breaks than before the pandemic.



Bob the bus driver needs a way to remain safe and healthy so that they can

Bob needs to make sure everyone is carrying a vaccination certificate while travelling so they can ensure passengers safety

Bob the bus driver and passengers need a way to make sure everyone on the bus follow rules so they can use the bus safely

Bob needs frequent temperature check

Bob the bus driver needs to ensure that passengers should maintain social distancing and follows covid-19 protocols so they can make sure everyone is travelling in a safe environment

Bob needs get tested if he gets any symptoms of covid

Need to make passengers feel safe

Bob needs to make sure that he & his family gets vaccinated



Need to know how many passengers is expected

Need to keep the bus less crowded

Bob the bus driver needs a way to make sure the bus is not overcrowded so that they can make sure the passengers have enough room for social distancing.

Bob the bus driver needs a way to make sure the bus is not overcrowded

AI Intents



<https://www.ibm.com/design/thinking/page/toolkit/activity/ai-essentials-intent>

1. With your group, pick TWO AI intents to help your users.
2. Brainstorm individually how you can accomplish your AI intents.
3. Group/cluster ideas into themes and label each group.
4. Vote on the best label/theme (2-3 votes for each group member).
5. Discuss which ideas are most important for users & business.

GROUP ACTIVITY (15 min)

AI Intents



- **Enrich your interactions**

Understand and communicate with customers and employees using natural language, responding to their needs with tailored dialogue and personalized experiences.

- **Anticipate and pre-empt disruptions**

Monitor your systems and equipment at all times so that can identify and address potential issues before they become bigger, more expensive problems.

- **Recommend with confidence**

Make more confident, targeted recommendations by using AI to evaluate a broad set of information based on an understanding of the parameters that are important to you.

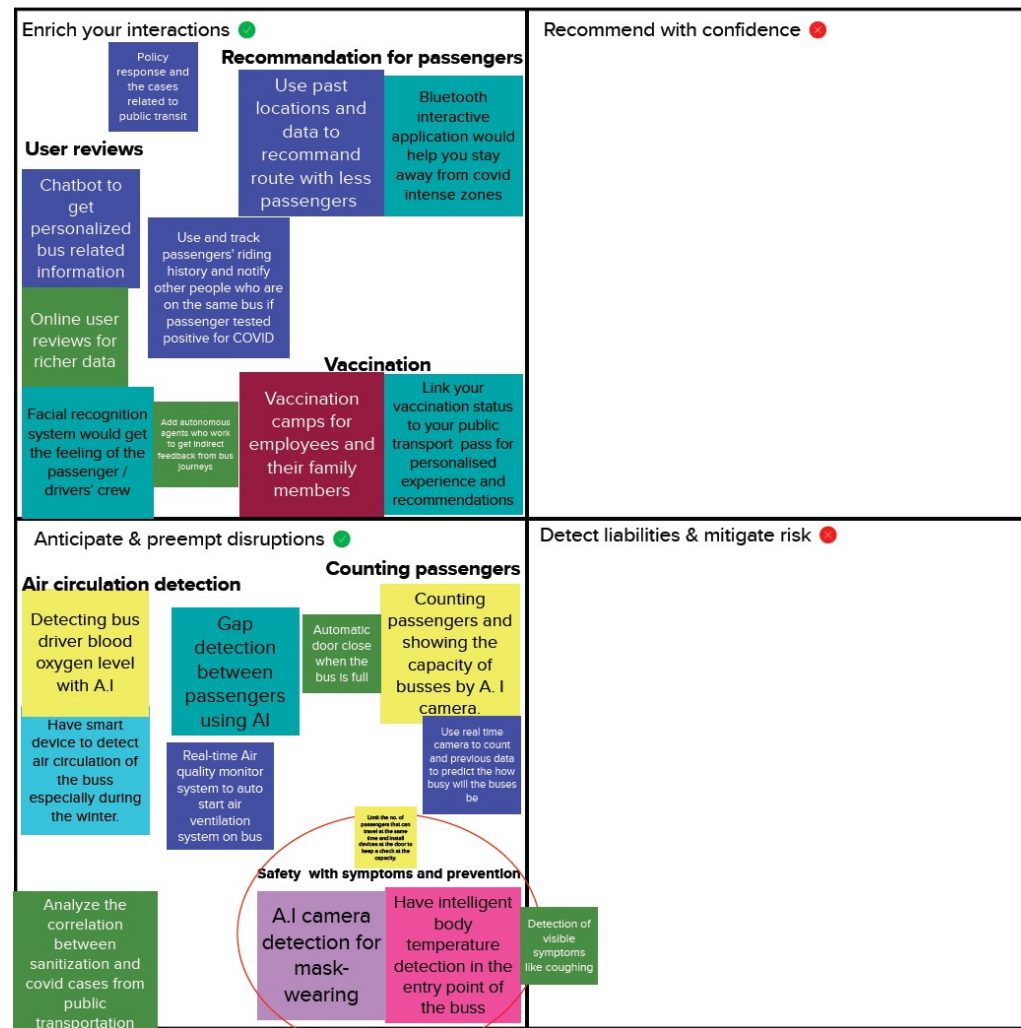
- **Detect liabilities and mitigate risk**

Use AI's understanding of the written word to identify risks to your company, particularly in terms of regulatory compliance.

AI Intents



AI INTENTS



AI Big Idea vignettes



<https://www.ibm.com/design/thinking/page/toolkit/activity/big-idea-vignettes>

VIDEO [01:47]

1. Copy a Needs Statement or Pain Point from As-Is-Scenario Map
2. Generate many ideas (experiences) with your solution, not features of your implementation. Eg: "Its kind of like..."
3. Cluster similar ideas. Name clusters. Discuss ideas and pick one(s) you want to advance.

GROUP ACTIVITY (15 min)

AI Big Idea vignettes

BIG IDEA VIGNETTES

Bob the bus driver needs to ensure that passengers should maintain social distancing and follows covid-19 protocols so they can make sure everyone is travelling in a safe environment

Keep the protocols respect

Announcement to remind passengers regarding the protocols

Add agents responsible of handling out the disrespect of the protocols

Distanciation

Make sure social distancing is respected

Blocking or removing seats for distanciation

Spaced out seat



Real-time monitoring

keeping a check on the no. of passengers in order to avoid the overcrowding.

Air quality monitoring

Air sanitisation during travel

Measure the air composition inside the bus

Mask monitoring

Symptoms detection

BoB should have access to high-level technology to protect people.

Live Covid detection

Summary of live passengers with symptoms and temperature

Body temperature check during on-boarding

Availability of rapid PCR test kit inside the vehicle

We can maybe detect symptoms of passengers

Assignment 4: Homework



To-Be-Scenarios & Data Sources